

Name _____

1 D Motion Test

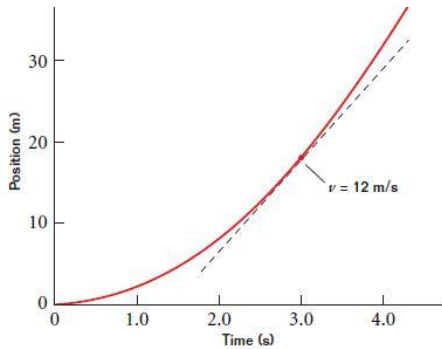
Multiple Choice - 2 pts each - Write the correct letter in the blank.

- _____ 1. An ant starts at position 2 cm and walks to position 25 cm. The ant's displacement is ____.
- 25 cm to the right
 - 2 cm to the left
 - 23 cm to the right
 - 20 cm to the left
- _____ 2. The change in position of an object is which of the following terms?
- Velocity
 - Displacement
 - Speed
 - Distance
- _____ 3. Negative displacement indicates ____.
- The movement is the the right
 - The movement is to the east
 - The movement is to the north
 - The movement is to the west
- _____ 4. Velocity is defined as ____.
- Change in position over change in time
 - Displacement over time interval
 - $\Delta x / \Delta t$
 - All of the above
- _____ 5. What is the velocity of an object at rest?
- 0.0 m/s
 - 1.0 m/s
 - 9.8 m/s
 - 9.81 m/s^2
- _____ 6. A dolphin swims 1.85 km/hr. How far does the dolphin swim after 36 minutes?
- 630 m
 - 3700 m
 - 2500 m
 - 1100 m

_____ 7. Acceleration is ____.

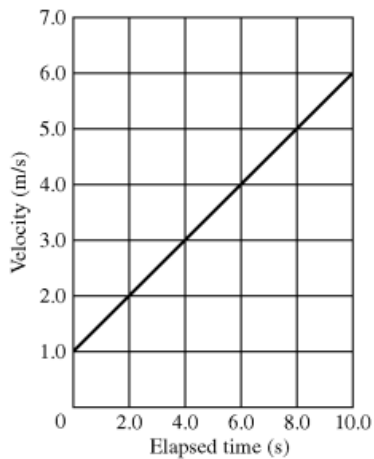
- a. Displacement
- b. The rate of change of displacement
- c. Velocity
- d. The rate of change of velocity

_____ 8. What does the following graph represent about velocity?



- a. Constant velocity
- b. That velocity is not constant
- c. Negative velocity
- d. Velocity of zero

_____ 9. What does the following graph represent about acceleration?



- a. Constant acceleration
- b. That acceleration is not constant
- c. Negative acceleration
- d. Acceleration of zero

_____ 10. What is the significance of negative acceleration, when an object's velocity is positive?

- a. The object is speeding up
- b. The object is slowing down
- c. The object is at rest
- d. None of the above

_____ 11. Acceleration due to gravity is also called ____.

- a. Negative velocity
- b. Displacement
- c. Free-fall acceleration
- d. Instantaneous velocity

____ 12. Objects that are in free fall toward the Earth _____.

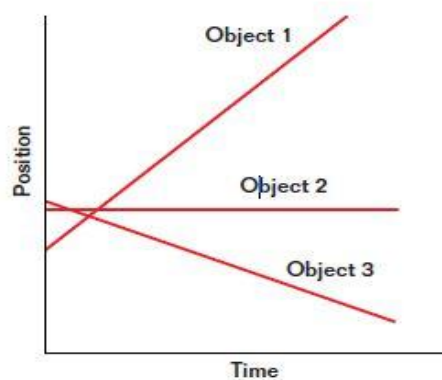
- a. Speed up as they approach the ground
- b. Slow down as they approach the ground
- c. Move at a constant velocity as they approach the ground
- d. Slow down and then speed up

____ 13. In a scenario without air resistance, what would hit the ground first if dropped from the top of the press box - a feather or metal bolt?

- a. Feather
- b. Metal bolt
- c. Hit the ground at the same time
- d. None of the above

____ 14. Which line represents an object at rest?

- a. Object 1
- b. Object 2
- c. Object 3
- d. All of the above



____ 15. What is missing on the graph in question 14?

- a. Scale
- b. Title
- c. Units
- d. All of the above

Matching - 2 pts each - Match the symbol with what it represents.

____ 16. Position

a. v_f

____ 17. Velocity

b. a_g or g

____ 18. Interval or change

c. x

____ 19. Time

d. v

____ 20. Acceleration

e. v_i

____ 21. Acceleration due to gravity

f. a

____ 22. Initial Velocity

g. t

____ 24. Final Velocity

h. Δ

Matching - 2 pts each - Match the unit with the symbol.

- | | |
|-------------|-------------------|
| _____ 25. x | a. m/s^2 |
| _____ 26. v | b. s |
| _____ 27. t | c. m |
| _____ 28. a | d. m/s |

1 D Motion Test
Steps to Work Problems

*If helpful, draw a picture that represents the situation

1. Given: Write down each number with a unit and identify which variable it represents using a letter/symbol
Symbol = number unit
2. Unknown: Identify the unknown variable
Symbol = ?
3. Equation: Decide which formula to use and copy it from the equation sheet.
4. Substitute: Plug in the numbers and the variable for what you are looking for.
5. Solve: Use algebra to solve for the unknown. Clearly write the answer with the correct unit.

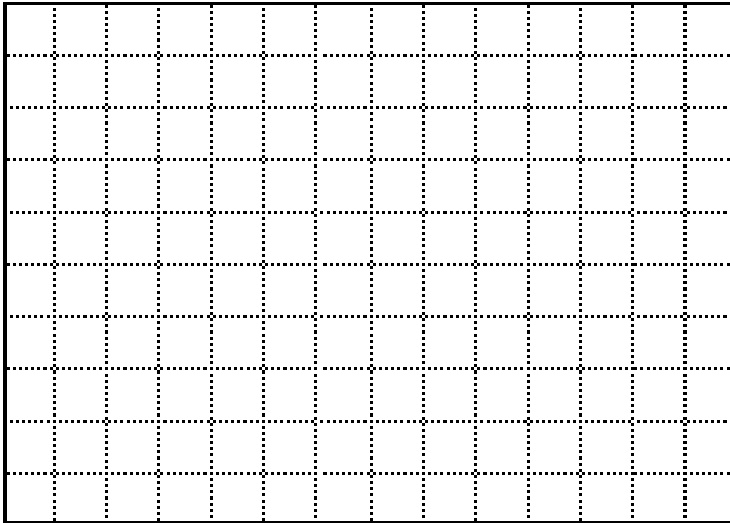
Remember to know what each letter/symbol stands for and what units are used.

Problems & Short Answer - points indicated

For the math problems remember to follow all the problem solving steps to receive full credit.

29. Construct a line graph of position versus time using the data in the table below. Be sure to include a title, axes labels, and units on the graph. (5 pts)

Position (m)	Time (s)
1.0	2.0
2.0	4.0
3.0	6.0
4.0	8.0
5.0	10.0



- a. What value is represented by the slope of the graph? (1 pts)
- b. Calculate the slope of the graph. (2pt)
- c. Identify your independent and dependent variable in the above data? (2 pts)

30. A bicyclist begins pedalling with a velocity of 12.0 m/s. He accelerates at a constant rate of 1.20 m/s². What is the final velocity after the bicyclist has traveled 200 m? (5 pts)

31. The Sears Tower in Chicago is 443 m tall. Joe wants to set the world's stair climbing record and runs all the way to the roof of the tower. If Joe's average upward velocity is 0.60 m/s, how long will it take Joe to climb from street level to the roof of the tower? (5 pts)

32. A cross-country skier with an initial forward velocity of $+4.42 \text{ m/s}$ accelerates uniformly at -0.75 m/s^2 until she comes to a stop.

a. How long does it take the skier to come to a stop? (5 pts)

b. What is the skier's displacement in this time interval? (5 pts)

33. A fighter jet lands on an aircraft carrier's flight deck. Although the deck is 300 m long, most of the jet's acceleration occurs within a distance of 42.0 m. If the jet's velocity is reduced uniformly from $+42.5 \text{ m/s}$ to 0 m/s as it moves through the 42.0 m, what is the jet's acceleration? (7 pts)

34. A tennis ball is released from rest at the top of the Simmons Bank Building in downtown Little Rock. Disregarding air resistance, calculate the displacement of the tennis ball after 1.00, 2.00 and 3.00 seconds. (8 pts)